

Almanac 47 for the NumWorks

Python version: the ALMANAC page and the horizon chart — user manual

Two graphic views of the C47 suite, drawn with the NumWorks's own graphics module (kandinsky, keys with ion): the “vintage” ALMANAC page and the horizon chart. The numbers come from nav.py and navdata.py — the same series and methods as the C47 programs (the Sun from the full VSOP87 series, valid 2000–2050, nutation, the 58 navigation stars, sunrise, twilight, Moon phase). On a PC every value on the screens was compared with the text page of nav.py and with the C47 ALMANAC screen: identical.

The Sun and the stars only: nav.py has no Moon position and no planets (the Moon phase and age are shown). The table has the Sun and the brightest stars higher than 10°, as on the C47.

It supports, and does not replace, the Nautical Almanac. Always cross-check the values.

Copyright © 2026 Victor Valdes. Free software under the GNU General Public License v3.0 or later. It comes with NO WARRANTY. Written with the help of AI (Claude, by Anthropic); the results were checked against the C47 version, which was checked against JPL Horizons, USNO data and the Nautical Almanac.

1. Files

File	From	What it is
nav.py, navdata.py	python/	the almanac computation and data (also a text menu)
nwlib.py	python/numworks/	text (the C47 5×7 font), symbols, keys, main loop
almview.py	python/numworks/	starts on the ALMANAC page
skyview.py	python/numworks/	starts on the horizon chart

2. Copy to the calculator

- **Epsilon (official):** on my.numworks.com create each script (Python → new script, paste the file, same name), then “Send to my calculator” with the calculator on USB (Chrome or Edge). The online simulator there runs them too.
- **Upsilon / Omega:** the same five files, with their web installer's script manager or the workshop.
- Open almview.py (or skyview.py) in the Python app and choose **Execute script**.

3. Inputs

The program asks in the terminal, like the nav.py menu:

Input	Format	Example
Year, Month, Day	numbers	2026, 9, 26
UT	hh:mm or hh:mm:ss	14:57
Lat, Lon	“25 20” = 25° 20′, “25.5” = decimal degrees, “-” for S / W	25 20, 55 12

- **Time is UT (UT1):** for a sight timed in UTC add DUT1 (under 0.9 s) — as in the almanac.
- **Signs:** latitude N +, S −; longitude E +, W −.

4. Keys

Key	Action
UP / DOWN	one hour later / earlier
OK or EXE	switch ALMANAC ↔ HORIZON
LEFT / RIGHT	HORIZON: the previous / next body for the top line (shown in red)
BACK	exit

5. The views

The previews were made on a PC with stand-ins for the kandinsky module (the same scripts; the calculator's own font may make text a little wider or narrower).

26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0

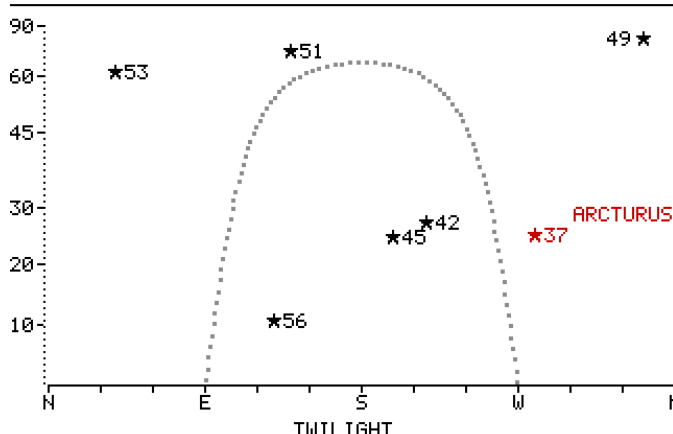
BODY	GHA	DEC	HC	ZN
ARIES	229 40.8			
☉ SUN	46 25.7 S 1 24.5	-11 06.4	273.7	
★37 ARCTURUS	15 27.7 N 19 02.7	24 59.7	280.2	
★49 VEGA	310 13.1 N 38 48.8	75 46.1	342.6	
★51 ALTAIR	291 39.3 N 8 56.5	69 23.0	140.4	
★42 ANTARES	341 55.1 S 26 29.5	27 00.7	217.3	
★56 FOMALHAUT	244 53.4 S 29 28.7	10 35.9	130.0	
★53 DENEK	279 05.4 N 45 22.8	61 13.8	039.3	
★45 SHAULA	325 49.4 S 37 07.5	24 29.0	198.3	

NAUT TWI 01:20 15:01 MOON 100% ● FULL
RISE/SET 02:09 14:12 AGE 15.5 DAYS
MER PASS 08:11 SD 16.0

UP/DOWN 1 HOUR OK HORIZON BACK EXIT

DOES NOT REPLACE THE NAUTICAL ALMANAC

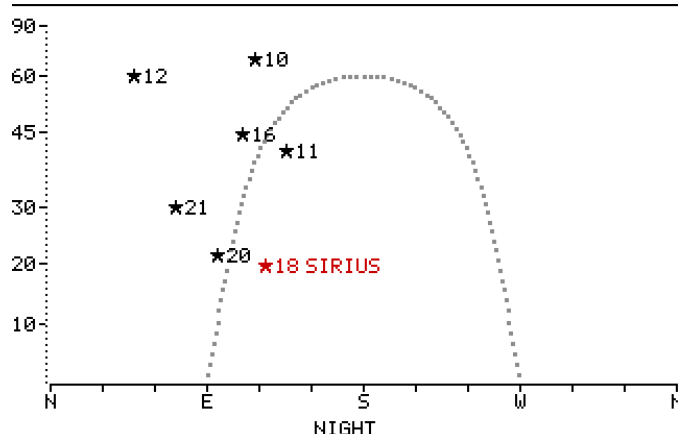
26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0



ALMANAC: date, UT, DR; GHA Aries; BODY, GHA, DEC, HC, ZN of the Sun and the stars; naut. twilight, rise/set, meridian passage, Sun SD, Moon phase and age. A negative Hc is white on black. 26 Sep 2026 14:57 UT, 25° 20' N 055° 12' E.

HORIZON: Hc up (sine scale), Zn across, the celestial equator dotted, the Sun and the stars with their numbers; the top line shows ZN and HC of the selected body. Same time and place.

23-09-2026 03:00 UT DR N 30 00.0 E 0 00.0



26-09-2026 14:57 UT DR N 25 20.0 E 55 12.0

BODY	GHA	DEC	HC	ZN
ARIES	229 40.8			
☉ SUN	46 25.7 S 1 24.5	-11 06.4	273.7	
★37 ARCTURUS	15 27.7 N 19 02.7	24 59.7	280.2	
★49 VEGA	310 13.1 N 38 48.8	75 46.1	342.6	
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NAUT TWI 01:20 15:01 MOON 100% ● FULL
RISE/SET 02:09 14:12 AGE 15.5 DAYS
MER PASS 08:11 SD 16.0

UP/DOWN 1 HOUR OK HORIZON BACK EXIT

DOES NOT REPLACE THE NAUTICAL ALMANAC

HORIZON at night: 23 Sep 2026 03:00 UT, 30° N 0° E (the Sun below the horizon).

6. The text menu

nav.py alone (Execute script, or **menu()** in the shell) is the text version: 1 Sun, 2 Star, 3 Rise/Set, 4 Moon phase, 5 Almanac (the same page as the ALMANAC view, as text).

7. Memory and size

- The five files are about 26 KB. The star data of navdata.py take about 15 KB of the Python heap.
- Epsilon 19 or later (64 KB heap) or Upsilon / Omega (about 100 KB): fine. Epsilon before 19 (32 KB heap): close to the limit — if you get MemoryError, update Epsilon or use Upsilon, and do not import other scripts in the same session.

8. Limitations

- **Series only, no almanac tables.** The Sun comes from the truncated VSOP87 series: valid 2000–2050, better than 0.1'. The tables TBL_1 / TBL_5 of the C47 and Free42 are not read (they mainly serve the Moon and the planets, which this version does not compute).
- **No Moon position and no planets:** only the Moon phase and age. For the Moon and the planets use the C47, Free42 / DM42n (NAVFULL) or the Python version on the PC (python/native).
- **ΔT fixed** at 69.2 s (TT – UT1), as in the C47 series; the input time is UT1 (add DUT1 to UTC).

- **Two views only:** ALMANAC and HORIZON (no split sky, all-sky, animation or star pages).
- **Memory:** on Epsilon before 19 (32 KB heap) the program is close to the limit (section 7); the almanac tables (1 year is about 8,400 numbers) could not fit in any NumWorks Python heap.

9. Status

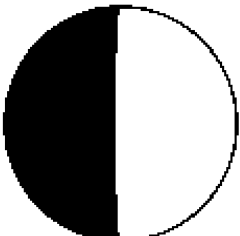
- Not yet tried on a real NumWorks. Checked on a PC (stand-in module) and in a MicroPython interpreter.

10. MOON47: the Moon phase

MOON47 is a separate program, the Moon's phase page without NAV, INIT or the tables:

python/numworks/moon47.py (about 10 KB, one file). Copy it to the calculator as moon47.py and run it. The NumWorks has no clock: it asks year, month, day and UT.

03-10-2026 12:00 UT



LAST QUARTER
LIT 51% AGE 22.4 DAYS
HP 59.2' SD 16.1'

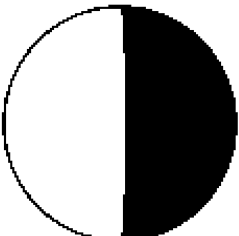
NEXT PHASES (UT)
LAST QUARTER 03-10 13:25
NEW MOON 10-10 15:50
FIRST QUARTER 18-10 16:13
FULL MOON 26-10 04:13

○ ○ ○ ● ● ● ● ● ○

AS SEEN FROM THE NORTH

OK NORTH/SOUTH BACK EXIT

03-10-2026 12:00 UT



LAST QUARTER
LIT 51% AGE 22.4 DAYS
HP 59.2' SD 16.1'

NEXT PHASES (UT)
LAST QUARTER 03-10 13:25
NEW MOON 10-10 15:50
FIRST QUARTER 18-10 16:13
FULL MOON 26-10 04:13

○ ○ ○ ● ● ● ● ● ○

AS SEEN FROM THE SOUTH

OK NORTH/SOUTH BACK EXIT

3 Oct 2026 12:00 UT, from the north

after OK: from the south

- **The page:** the Moon as a disc (the lit part filled), the phase name, % lit, age in days, HP and SD, the next new Moon, first quarter, full Moon and last quarter (date and time UT), and the eight phase symbols with today's one inverted. The phase does not depend on where you are; the latitude only turns the picture.
- **Keys:** OK or EXE turns the picture to the view from the south and back; BACK ends.
- **Accuracy** against the full series of NAV, 2000–2050: phase times and age within 4 minutes (most within 1), HP 0.03', SD 0.01', % lit 0.03 %: the 20 largest terms of the Moon (Meeus, ch. 47), the same formulas as MOON47 on the C47 (python/moon47.py; tests/test_moon47.py).